



Department of Education

STEM Education
National Schools of Excellence
PHYSICS Teacher's Guide
Grades 12
Term 3

<u>DOMAIN</u>	<u>PHYLA</u>
Science	Physics
Technology	
Mathematic	

Issued free to schools by the Department of Education

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SECRETARY'S MESSAGE

This PHYSICS Syllabus is to be used exclusively by teachers of PHYSICS to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM PHYSICS Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in PHYSICS before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating PHYSICS as a fundamental field of science.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of PHYSICS is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of PHYSICS to human lives, including good health and environment.

In order to ensure Teachers, discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for PHYSICS to be used for the Grade 11 STEM Education Program in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

The PHYSICS Syllabus introduced to the Schools of Excellence as a critical component of the STEM Education is mostly concerns

Teaching PHYSICS requires laboratory sessions as an essential part of the teacher guide. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of PHYSICS.

The Grade 11 PHYSICS Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since PHYSICS is a highly specialized subject students undertaking PHYSICS are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of PHYSICS through the process of imparting basic PHYSICS knowledge and skills to student

THEMA	MODULE	TOPICS
ARTIFICIAL INTELLIGENCE	12.1 Data and Artificial Intelligence	1. Introduction to Artificial Intelligence 2. Machine Learning 3. Neural Networks 4. Data Mining
ROBOTICS	12.1 Robotics and Electronics	1. Introduction to Robotics and Electronics 2. Control Systems 3. Circuit Boards

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

The PHYSICS was introduced as a Bridging Course to the Schools of Excellence. The syllabus introduced the students to the PHYSICS Domain and expects them to develop an appreciation of the discipline in a broader context. In this teacher guide students are introduced to advanced concepts in PHYSICS as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM Domain

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, PHYSICS and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with Physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the PHYSICS design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and PHYSICS and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in PHYSICS design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. PHYSICS utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the PHYSICS Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in PHYSICS. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM PHYSICS CONTENT OVERVIEW – GRADE 12, TERM 3

Table 1.1: Sequencing & Benchmarking STEM PHYSICS Content Standards

Setting STEM PHYSICS Content Standards for Progressive Learning and Assessment National STEM Assessment			
DOMAIN: PHYSICS			
Thema: Artificial Intelligence		Broad Content Standard:	
Module 12.1: Data and Artificial Intelligence		12.1a: Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining 12.1b: Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence	
Topic	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Artificial Intelligence	12.1.1a: Define Artificial Intelligence 12.1.1b: List branches of AI and explain their role in the development of AI 12.1.1c: Provide example applications of Artificial Intelligence in Health Care, Robotics and Navigation	Students' ability to understand the knowledge and skills of: - Machine Learning - Neural Networks - Data Mining	Group Presentation/Research + Written Quiz + Written Test + Assignment
2. Machine Learning	12.1.2a: Research on the different type of Machine Learning 12.1.2b: Conduct a Presentation based on their research on the different types of Machine Learning 12.1.2c: Define Machine Learning 12.1.2d: Define Deep Learning		
3. Neural Networks	12.1.3a: Relate Neurobiology to Artificial Intelligence Neural Networks 12.1.3b: Explain the theoretical foundations of Neural Networks 12.1.3c: Research and present explanation and an example on Supervised Training in Neural Networks 12.1.3d: Research and present explanation and an example on Unsupervised Training in Neural Networks 12.1.3e: Research and present explanation and an example on Self-Supervised Training in Neural Networks		
4. Data Mining	12.1.4a: Define and explain data mining 12.1.4b: Describe the processes involved in data mining 12.1.4c: Explain the techniques of data mining 12.1.4d: Conduct Research on the different applications of data mining		

Thema: Robotics		Broad Content Standard:	
Module 12.1: Robotics and Electronics		12.1a: Assemble a PCB 12.1b: Design a PCB 12.1c: Assemble a robot	
Topic	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Robotics and Electronics	12.2.1a: Define and explain what robotics is 12.2.1b: Describe the purpose of robotics 12.2.1c: Highlight and explain the main components of a robot 12.2.1d: Describe the main functions of a robot 12.2.1e: Apply knowledge of robotics to its applications	Students' ability to understand the knowledge and skills of: • Robotic Control Systems • Circuit Boards • Assembling and design a robot	Group Presentation/Research + Written Quiz + Written Test + Assignment
2. Control Systems	12.2.3a: Explain control systems and give examples 12.2.3b: Describe how control systems work 12.2.3c: Design a simple control system		
3. Circuit Boards	12.2.3a: Define and explain circuit boards 12.2.3b: Describe the functions of circuit boards 12.2.3c: Explain the different types of circuit boards 12.2.3d: Highlight the importance of circuit boards 12.2.3e: Explain how circuit boards work 12.2.3f: Apply theory and create an actual circuit board		

Table 1.2: Mapping Content Progression by Modules

Thema/Module	Content Standard	12.1	12.1
ARTIFICIAL INTELLIGENCE / 12.1 Data and Artificial Intelligence	12.1a: Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining 12.1b: Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence	✓	
ROBOTICS / 12.1 Robotics and Electronics	12.1a: Assemble a PCB 12.1b: Design a PCB 12.1c: Assemble a robot		✓

THEMA: ARTIFICIAL INTELLIGENCE (AI)

Introduction

Artificial Intelligence (AI) refers to the intelligence demonstrated by machines rather than natural intelligence expressed by humans or animals. This mimicked intelligence by machines refers to the cognitive functions, problem-solving methodologies and behavior associated particularly with humans and this is the common perception that is expressed by most people. In this sense, AI gleaned its inspiration from the human brain.

In 1956, the academic discipline of Artificial Intelligence was founded and established by a group of experts from different disciplines (including Mathematics, Engineering, Psychology and Political Science) that sought to discover the possibilities of creating an artificial brain. As such, AI research had explored and rejected a variety of methodologies, including brain simulation and problem-solving modeling. In the first two decades of the twenty-first century, highly mathematical and statistical modelling dominated the subject and created useful applications such as search engines, recommendation systems, and self-driving cars. As technologies improved with research and resources to develop more capable machines became readily available, actions that were previously called "intelligent" were gradually being excluded from AI definitions.

AI technologies are currently used for a variety of applications including robotics for space and underwater exploration, an automated assembly line of robots in large manufacturing industries, and auto correct and face recognition on smartphones. As AI research and technology improves, it is becoming more accessible and integral to the common every-day human being and could become a predominant tool in resolving almost all of humanities problems. This would also suggest that there will be a lesser dependency on humans themselves or human intervention to solve our own problems. This has planted doubts into AI skeptics, developers, and companies on the uncertainties of AI technologies with concerns of it being a possible threat to human survival which is raising many controversial debates today. However so, it is an undisputable fact that AI has improved and aided in the design of the modern world as we know it today.

It is important that students are to be introduced to the concept of AI in preparation of its vast applications in the future. With this introduction, students should be thinking about how they can become creators and innovators in the field of AI and AI development, and not only end-users.

MODULE 12.1: DATA AND ARTIFICIAL INTELLIGENCE

4 weeks

The field of Artificial Intelligence is deeply rooted in using data and information. Based on how its model is created, the data is manipulated to develop “reasoning” that allows the AI to perform its intended function. This is the general field of AI that then diverges into a variety of sub-fields that are categorized according to certain distinct objectives and methods. In this *Thema: Artificial Intelligence*, we will explore and discuss some of these sub-fields and focus on some of the current practices used in AI.

Linking Modules

- Technology / Grade 12 / Thema Artificial Intelligence / Module 12.1
- Engineering / Grade 12 / Thema Artificial Intelligence / Module 12.1
- Chemistry / Grade 12 / Thema Artificial Intelligence / Module 12.1

Broad Content Standard

12.1a: Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining

12.1b: Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence

Content Benchmarks

Students’ ability to understand the knowledge and skills of:

- Machine Learning
- Neural Networks

Data Mining

Topic 1: Introduction to Artificial Intelligence

Performance Indicators

Students are expected to:

12.1.1a: Define Artificial Intelligence

12.1.1b: List branches of AI and explain their role in the development of AI

12.1.1c: Provide example applications of Artificial Intelligence in Health Care, Robotics and Navigation

Activity 1: Exercises

Prepare a Worksheet containing exercises with questions pertaining to the concepts taught in the topic.

Resources: NIL

(Refer to TRB)

Topic 2: Machine Learning

Performance Indicators

Students are expected to:

- 12.1.2a:** Research on the different type of Machine Learning
- 12.1.2b:** Conduct a Presentation based on their research on the different types of Machine Learning
- 12.1.2c:** Define Machine Learning
- 12.1.2d:** Define Deep Learning

Activity 1: Assignment

Title: Types of Machine Learning

Prepare an assignment for students to do research and answer the following points.

- List the different types of machine learning techniques.
- Explain each of these types.
- Give examples for each.

Resources: NIL

(Refer to TRB)

Activity 2: Group Discussion

- 1) Group Students
- 2) Task each group to explain in their own understanding what Supervised Learning, Unsupervised Learning and Reinforcement Learning is. Each group must also give examples regarding the applications of the types of Machine Learning above.
- 3) Each Group to have representatives present their discussions to the class.

Resources: Marker, White Board, Chalk, Chalk board

(Refer to TRB)

Activity 3: Exercises

Prepare a Worksheet containing exercises with questions pertaining to the concepts taught in the topic.

Resources: NIL

(Refer to TRB)

Topic 3: Neural Networks

Performance Indicators

Students are expected to:

- 12.1.3a:** Relate Neurobiology to Artificial Intelligence Neural Networks
- 12.1.3b:** Explain the theoretical foundations of Neural Networks
- 12.1.3c:** Research and present explanation and an example on Supervised Training in Neural Networks
- 12.1.3d:** Research and present explanation and an example on Unsupervised Training in Neural Networks
- 12.1.3e:** Research and present explanation and an example on Self-Supervised Training in Neural Networks

Activity 1: Group Presentation

- 1) Group Students
- 2) Each group is to select a topic below to research on and prepare for Class Presentation on Neural Networks. Each student must include in their research a case study or example of research topic.
 - a. Explain how Neurobiology compares to Neural Networks.
 - b. How does Supervised Training in Neural Networks Work?
 - c. How does Unsupervised Training in Neural Networks Work?
 - d. How does Self-supervised Training in Neural Networks Work?

Resources: Projector, Speakers

(Refer to TRB)

Topic 4: Data Mining

Performance Indicators

Students are expected to:

- 12.1.4a:** Define and explain data mining
- 12.1.4b:** Describe the processes involved in data mining
- 12.1.4c:** Explain the techniques of data mining
- 12.1.4d:** Conduct Research on the different applications of data mining

Activity 1: Group Research and Presentation

- 1) Group Students
- 2) Each group is to select one of the research topics and explain in simple terms how Data Mining is used under that topic.
 - a. Financial Data Analysis
 - b. Retail Industry
 - c. Telecommunication Industry
 - d. Biological Data Analysis
 - e. Other Scientific Applications
 - f. Intrusion Detection

Resources: Projector, Speakers

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks	
12.1: Data and Artificial Intelligence	12.1a: Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining 12.1b: Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence	1. Introduction to Artificial Intelligence	Written Quiz	Project + Written Test
		2. Machine Learning		
		3. Neural Networks	Written Quiz	
		4. Data Mining		

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1 and 2	Written Quiz	Define Artificial Intelligence	2.5
		List branches of AI and explain their role in the development of AI	2.5
		Provide example applications of Artificial Intelligence in Health Care, Robotics and Navigation	5
		Research on the different type of Machine Learning	5
		Conduct a Presentation based on their research on the different types of Machine Learning	2.5
		Define Machine Learning	2.5
Subtotal			20
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	30
		Correctly answers short answer questions	30
		Subtotal	80
Module TOTAL			100

THEMA: ROBOTICS

Introduction

Robotics is a specific yet versatile field of study and is a branch of AI. It utilizes robots (hence the name robotics) that are endowed with the equivalence of human senses including vision, touch, and the ability to sense surrounding conditions such as temperature and pressure to perform simple to complicated tasks.

Robots possess the capabilities to exceed human limitations such as to work longer hours, be accurate and precise in calculations and actions, and to work in hazardous conditions. Much like AI, robotics has a widespread of applications including manufacturing industries, medical healthcare, and military purposes for security and defense intelligence.

There are robots that are designed to accomplish pre-programmed instructions and there are some programmed to make simple decisions. Current robotics research is focused on improving this decision-making process for robots to become self-sufficient; allowing them to make decisions in unconstructed environments.

AI and machine learning algorithms have grown more efficient because of advancements in computing, data transfer, storage, and computer power, making it easier to encode programs for robots, design innovative uses for applications, and minimize energy consumption. Additionally, the improvements in sensor technology and material science improve the hardware of robotics to interface with AI capabilities, ultimately improving the field of robotics.

Although much focus is directed towards product technology (e.g., smartphones, commercial jets, automobiles, solar panels, etc.), process technology (which involves the usage of machines to improve how a good or service is produced, enabling automation) will influence significant changes that will later redefine the scope of modern technology.

MODULE 12.1: ROBOTICS AND ELECTRONICS

4 weeks

Robotics is a multidisciplinary field combining computer science and engineering. Robotics is the study of the design, manufacture, operation, and use of robots. Robotics aims to create devices that can support and help humans. Mechanical engineering, electrical engineering, information engineering, mechatronics, electronics, bioengineering, computer engineering, control engineering, software engineering, and mathematics are just a few of the subjects that robotics encompasses.

Robots can be employed in a variety of scenarios and for a variety of objectives, but many are now used in hazardous areas (such as radioactive material inspection, bomb detection and deactivation), manufacturing operations, or in situations where humans are unable to live (e.g., in space, underwater, in high heat, and clean up and containment of hazardous materials and radiation). Robots can take on any shape, but some are designed to seem like people.

This module introduces students to Robotics; it highlights the different aspects of the module electronics and describes the different applications of robotics in the real world.

Linking Modules

NIL

Broad Content Standard

12.1a: Assemble a PCB

12.1b: Design a PCB

12.1c: Assemble a robot

Content Benchmarks

Students' ability to understand the knowledge and skills of:

- Robotic Control Systems
- Circuit Boards

Assembling and design a robot

Topic 1: Introduction to Robotics and Electronics

Performance Indicators

Students are expected to:

12.2.1a: Define and explain what robotics is

12.2.1b: Describe the purpose of robotics

12.2.1c: Highlight and explain the main components of a robot

12.2.1d: Describe the main functions of a robot

12.2.1e: Apply knowledge of robotics to its applications

Activity 1: Student Task

Using the mentioned components in this section, students are to construct a diagram showing how these components work together in a robot.

Resources: NIL

(Refer to TRB)

Activity 2: Group Discussion

1. Group Students
2. Task each group to come up with an innovative solution using robots to solve a problem of their choice in Papua New Guinea.
3. Each group must choose a representative to present in class their chosen problem and their innovative solution using robotics, elaborating as clearly as they can.

Resources: NIL

(Refer to TRB)

Activity 3: Group Presentation

1. Group Students
2. Task each group to select one of the applications below and select a robot to discuss on.
 - a. Robots used in Industrial Setting.
 - b. Robots used in Research.
 - c. Robots used in Education.
 - d. Robots in space expeditions.
 - e. Robots used in deep sea exploration.
 - f. Robots used in mining industries.
3. Each group is to create a PowerPoint Presentation to show and explain to the class.

Resources: Projector, Speakers

(Refer to TRB)

Activity 4: Exercises

Prepare a Worksheet containing exercises with questions pertaining to the concepts taught in the topic.

Resources: NIL

(Refer to TRB)

Topic 2: Control Systems

Performance Indicators

Students are expected to:

12.2.3a: Explain control systems and give examples

12.2.3b: Describe how control systems work

12.2.3c: Design a simple control system

Activity 1: Video Learning

Title: Open and Closed Loop Systems in Robotics

Setup a Projector and have the students watch a video explaining open and closed loop systems in Robotics. (Encourage for visual learners). Link:

<https://youtu.be/f32SDx3C6sA>

Resources: Projector, Speakers

(Refer to TRB)

Activity 2: Class Exercise

Title: Closed Loop Control Systems

For this class exercise, students are to do their personal research at home and discuss with their teachers in class about the fundamentals of a closed loop control system.

Resources: NIL

(Refer to TRB)

Activity 3: Assignment

Title: Robotic Control Systems

For this assignment, each student is to do their research and answer the following points:

- List the two categories for the control systems of robots
- Explain each of them by highlighting the mechanics and physics involved in their design and functions

Resources:

(Refer to TRB)

Topic 3: Circuit Boards

Performance Indicators

Students are expected to:

12.2.3a: Define and explain circuit boards

12.2.3b: Describe the functions of circuit boards

12.2.3c: Explain the different types of circuit boards

12.2.3d: Highlight the importance of circuit boards

12.2.3e: Explain how circuit boards work

12.2.3f: Apply theory and create an actual circuit board

Activity 1: Practical or Project

Title: Assembling a PCB

Get Students to assemble a PCB.

Resources: Electronics Technician Tool Box

(Refer to TRB)

Activity 2: Practical or Project

Title: Designing of a PCB

Get students in groups and have them build a simple PCB and present the product at the end of the term. They are to explain the materials involved in designing the PCB and elaborate on its functions as well as the steps involved in the design process.

Resources: (Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks	
12.1: Robotics and Electronics	12.1a: Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining 12.1b: Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence	1. Introduction to Robotics and Electronics	Written Quiz	Presentation + Project + Written Test
		2. Control Systems		
		3. Circuit Boards	Written Quiz	

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1	Written Quiz	Define and explain what robotics is	2.5
		Define Electronics	2.5
		Describe the purpose of robotics	5
		Highlight and explain the main components of a robot	5
		Describe the main functions of a robot	2.5
		Apply knowledge of robotics to its applications	2.5
Subtotal			20
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	30
		Correctly answers short answer questions	30
		Subtotal	80
Module TOTAL			100

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules	Total Marks	% Weight of Modules
	6		
Theory Knowledge & skills – Test	Test (50)	/50	20
Application of knowledge – Assignments and projects	Assign.1 (30)	/30	30
End of Term Exam	(25 marks)	/25	50
Total marks for term one			100 marks

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Name s					Assessment Task				Total Score (/300)	Percentage Weight (%)				Achie vemen t Level [VHA, HA, SA,LA]
		Tests		Assignments				Project	Exam (/50)		T (20%)	A + P (30%)	Exam (50%)	Ttl CA	
		1	2	1	2	3	4	1							
1	Bill Joe	45	47	20	21	22	23	55	45	278	18.4	25.88	45	89.2 8	VHA
2	Cara Den	20	25	15	16	17	18	25	30	166	9	18.22	30	57.2 2	SA
3	Lea May	9	15	5	6	7	8	30	25	105	4.8	9.10	25	38.9	LA

Assessment Reporting

Thema Robotics / Module 12.1: Robotics and Electronics Broad Content Standard for Assignments / Practical 12.1a: Assemble a PCB 12.1b: Design a PCB 12.1c: Assemble a robot				
Performance Criteria	Robotics Concept (15 marks)	Design and assembling a robot (5 marks)	Answers all Relative Questions to Robot Design and Construct (5 marks)	25 marks
Very High Achievement	Accurately understands and explains the concept of robotics, its function and applications. (13-15)	Accurately understands how to assemble and design a PCB Board (5)	Accurately answers all relative questions to robot design and construct (5)	(21-25)
High Achievement	Partially understands and explains the concept of robotics, its function and applications. (10-12)	Partially understands how to assemble and design a PCB Board (4)	Partially answers all relative questions to robot design and construct (4)	(16-20)
Satisfactory Achievement	Fairly understands and explains the concept of robotics, its function and applications. (7-9)	Fairly understands how to assemble and design a PCB Board (3)	Impartially answers all relative questions to robot design and construct (3)	(11-15)
Low Achievement	Poorly understands and explains the concept of robotics, its function and applications. (4-6)	Poorly understands how to assemble and design a PCB Board (2)	Poorly answers all relative questions to robot design and construct (2)	(6-10)
Below minimum Achievement	Needs improvement in understanding and explaining the concept of robotics, its function and applications. (0-3)	Needs improvement in understanding how to assemble and design a PCB Board (1)	Needs improvement in answering all relative questions to robot design and construct (1)	(0-5)

Thema Robotics / Module 12.1: Robotics and Electronics Broad Content Standards for Test: 12.1a: Assemble a PCB 12.1b: Design a PCB 12.1c: Assemble a robot				
Performance Criteria	Correctly answers all long questions (25 marks)	Correctly answers all multiple choice (10 marks)	Correctly use concepts to design and propose solutions(15 marks)	50 marks
Very High Achievement	Accurately answers all long questions (21-25)	Accurately answers all multiple choice (9-10)	Accurately use concepts to design and propose solutions (13-15)	(41-50)
High Achievement	Partially answers all long questions (16-20)	Partially answers all multiple choice (7-8)	Partially use concepts to design and propose solutions (10-12)	(31-40)
Satisfactory Achievement	Fairly answers all long questions (11-15)	Fairly answers all multiple choice (5-6)	Fairly use concepts to design and propose solutions (7-9)	(21-30)
Low Achievement	Poorly answers all long questions (6-10)	Poorly answers all multiple choice (3-4)	Poorly use concepts to design and propose solutions (4-6)	(11-20)
Below minimum Achievement	Needs improvement in answering all long questions (0-5)	Needs answers all multiple choice (0-2)	Needs improvement in using concepts to design and propose solutions (0-3)	(1-10)

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